Budget Allocation for Household Needs

Team Members:

SE23UCAM014
SE23UCAM019
SE23UCAM025
SE23UCAM027

Contents

1 Introduction **2**

2 Optimization Problem **2**
2.1 Mathematical formulation 2
2.1.1 Decision variable 2
2.1.2 Objective Function 2
2.1.3 Constraints 2
2.2 Graphical solution 3

3 Big M Method **4**
3.1 Problem Statement 4
3.1.1 Decision Variables 4
3.1.2 Objective Function 4
3.1.3 Constraints 5
3.2 Standard Form Conversion and Big M Method Justification 5
3.3 Initial Simplex Table 6
3.4 Iterations 7
3.5 Final Optimal Solution 12

4 Advantages and Disadvantages of Methods Used **12**
4.1 Graphical Method 12
4.2 Big M Method 13
4.3 When to Use Each Method 14

5 Conclusion **14**

1 Introduction

Optimization is a fundamental concept in decision-making that involves selecting the best possible option from a set of alternatives to achieve a desired objective. Linear Programming Problem (LPP) is a widely used mathematical technique in optimization that helps in determining the optimal allocation of limited resources to maximize or minimize a linear objective function subject to constraints. In this project, we explore the application of LPP to optimize household budget allocation. By modeling the household expenses as decision variables and defining an objective function to maximize overall satisfaction or utility, we demonstrate how LPP can provide effective solutions to real-life resource allocation problems. The project covers both graphical and algebraic solution methods, including the Big M method, to solve LPPs with two and multiple variables respectively.

2 Optimization Problem

In this project, we will look at optimizing budget allocation for a households need. Suppose a household has a monthly budget of Rs. 50,000 and they want to allocate this resource into two essential categories - food and maintenance (electricity, water, gas etc.) Here, the aim would be to maximize the total utility or satisfaction derived from the spending of these categories. We will measure utility in utils. Each rupee spent on these categories provides a different level of utility.

2.1 Mathematical formulation

2.1.1 Decision variable

Let X and Y be the two decision variables.
X = amount spent on food (in Rs.)
Y = amount spent on maintenance (Rs.)

2.1.2 Objective Function

Assume that Rs. 1000 spent on food gives 8 utils of satisfaction and Rs. 1000 spent on maintenance gives 6 utils. The objective is to maximize utility/satisfaction.

$$\textbf{Maximize Z = 8X + 6Y}$$

2.1.3 Constraints

$$\begin{aligned} X + Y &\leq 50000 \\ X &\leq 30000 \\ Y &\leq 15000 \\ X &\geq 0 \\ Y &\geq 0 \end{aligned}$$

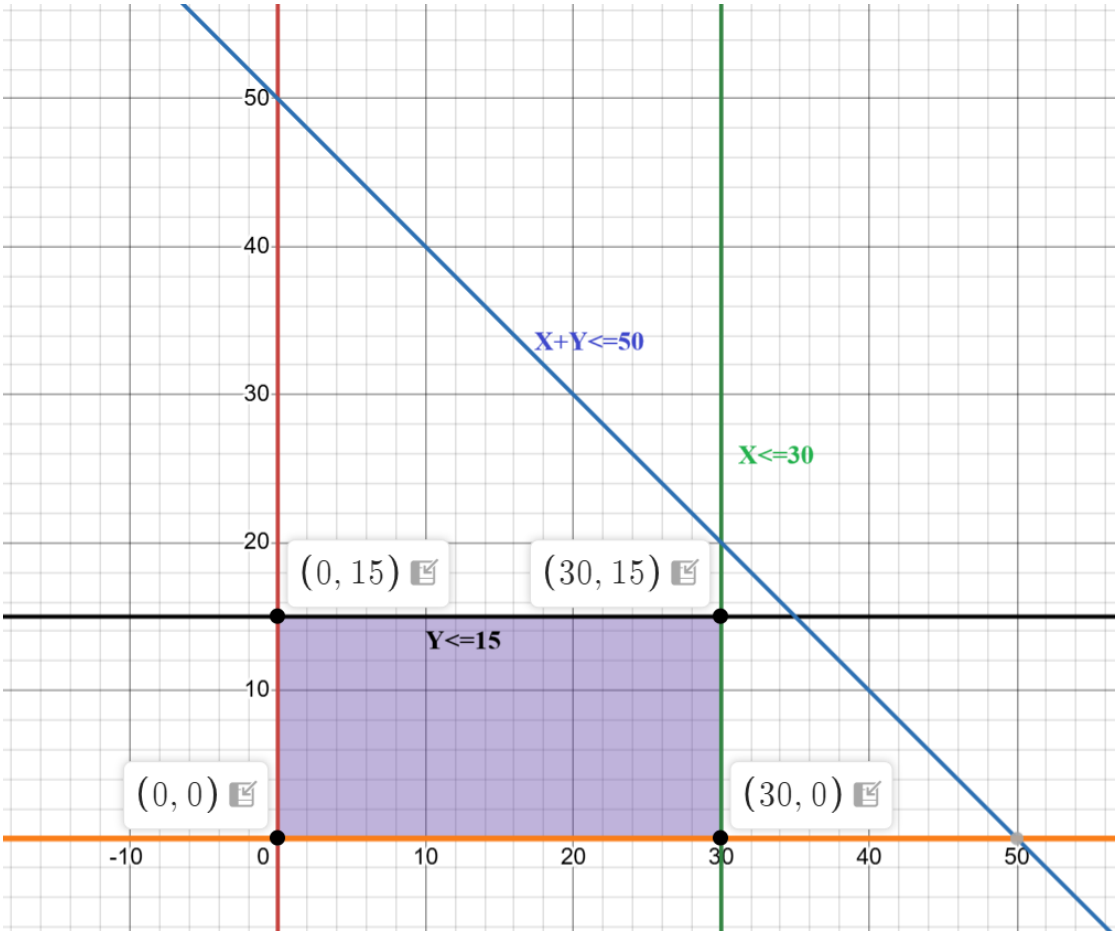

Figure 1: Graphical Solution

2.2 Graphical solution

Graph Key:

- X–axis: 1 unit = Rs. 1,000 (Food)
- Y–axis: 1 unit = Rs. 1,000 (Maintenance)

Feasible Region Vertices:

$$A = (0, 0),$$
$$B = (0, 15),$$
$$C = (30, 15),$$
$$D = (30, 0).$$

Since every Rs.1000 gives 8 utils satisfaction from food and 6 utils satisfaction from maintenance we input these points directly in objective function. Evaluation at Corner Points:
Z=f(x,y)

We now calculate Z values at points A, B, C, and D

$$\begin{aligned}
f(A) = f(0,0) &= 8 \cdot 0 + 6 \cdot 0 = 0, \\
f(B) = f(0,15) &= 8 \cdot 0 + 6 \cdot 15 = 90, \\
f(C) = f(30,15) &= 8 \cdot 30 + 6 \cdot 15 = 330, \\
f(D) = f(30,0) &= 8 \cdot 30 + 6 \cdot 0 = 240.
\end{aligned}$$

The maximum value of the objective function occurs at the point $(30, 15)$. This is when X= 30,000 and Y= 15,000. So, X+Y= 45,000 which is less than 50,000. This satisfies the constraint $X + Y \leq 50,000$ (i.e., total spending within Rs. 50,000). Hence, a household can achieve a maximum satisfaction of **330 utils** when they allocate **Rs. 30,000 on Food** and **Rs. 15,000 on Maintenance**. This allocation lies within the feasible region and utilizes the available budget most effectively to maximize overall utility.

3 Big M Method

3.1 Problem Statement

Now, for more than two variables let us consider a family consisting of 2 adults and 1 child with monthly income of Rs.50,000. They aim to allocate this budget across various essential and lifestyle categories in a way that maximizes their overall utility/satisfaction. Each category contributes differently to the family's satisfaction, and certain minimum and maximum spending constraints exist for some categories.

3.1.1 Decision Variables

Let

$$\begin{aligned}
X_1 &= \text{Amount spent on Rent} \\
X_2 &= \text{Amount spent on Groceries and Daily Essentials} \\
X_3 &= \text{Amount spent on Child's Education} \\
X_4 &= \text{Amount spent on Transport} \\
X_5 &= \text{Amount spent on Health} \\
X_6 &= \text{Amount allocated to Savings} \\
X_7 &= \text{Amount spent on Entertainment}
\end{aligned}$$

3.1.2 Objective Function

Maximize the total utility:

$$\textbf{Maximize Z} = 9X_1 + 10X_2 + 9X_3 + 6X_4 + 8X_5 + 10X_6 + 5X_7$$

The numerical coefficients (e.g., 9, 10, 6) represent the *utils of satisfaction* received per unit of currency (e.g., per Rs. 1000) spent in that category. For example, spending Rs. 1000 on groceries (X_2) yields 10 utils of satisfaction, while the same amount spent on entertainment (X_7) yields 5 utils.

3.1.3 Constraints

1. Minimum Spending Constraints:

$$X_1 \geq 10000 \quad -(1) \qquad \text{(Rent)}$$
$$X_2 \geq 7000 \quad -(2) \qquad \text{(Groceries)}$$
$$X_3 \geq 5000 \quad -(3) \qquad \text{(Education)}$$
$$X_5 \geq 3000 \quad -(4) \qquad \text{(Healthcare)}$$
$$X_6 \geq 3000 \quad -(5) \qquad \text{(Savings)}$$

2. Maximum Spending Constraints:

$$X_4 \leq 4000 - (6) \qquad \text{(Transport)}$$
$$X_7 \leq 5000 - (7) \qquad \text{(Entertainment)}$$

3. Total Budget Constraint:

$$X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 \leq 50000 - (8)$$

4. Non-negativity Constraints:

$$X_1, X_2, X_3, X_4, X_5, X_6, X_7 \geq 0$$

3.2 Standard Form Conversion and Big M Method Justification

The problem will be adapted to the standard lLPP by introducing slack, surplus, and/or artificial variables into each constraint to convert inequalities into equalities:

- **Constraint 1:** It has a "$\geq$" sign. Therefore, a surplus variable S_1 will be subtracted and an artificial variable A_1 will be added.
- **Constraint 2:** It has a "$\geq$" sign. Therefore, a surplus variable S_2 will be subtracted and an artificial variable A_2 will be added.
- **Constraint 3:** It has a "$\geq$" sign. Therefore, a surplus variable S_3 will be subtracted and an artificial variable A_3 will be added.
- **Constraint 4:** It has a "$\leq$" sign. Therefore, a slack variable S_4 will be added.
- **Constraint 5:** It has a "$\geq$" sign. Therefore, a surplus variable S_5 will be subtracted and an artificial variable A_4 will be added.
- **Constraint 6:** It has a "$\geq$" sign. Therefore, a surplus variable S_6 will be subtracted and an artificial variable A_5 will be added.
- **Constraint 7:** It has a "$\leq$" sign. Therefore, a slack variable S_7 will be added.
- **Constraint 8:** It has a "$\leq$" sign. Therefore, a slack variable S_8 will be added.

Since the problem contains artificial variables, we apply the **Big M method**. As this is a maximization problem, each artificial variable will be subtracted from the objective function and multiplied by a very large positive number M. This penalization ensures that the simplex algorithm eliminates artificial variables from the basis in the optimal solution.

Objective Function:

$$\begin{aligned}\textbf{Maximize Z} = {} & 9X_1 + 10X_2 + 9X_3 + 6X_4 + 8X_5 + 10X_6 + 7X_7 \\ & + 0S_1 + 0S_2 + 0S_3 + 0S_4 + 0S_5 + 0S_6 + 0S_7 + 0S_8 \\ & - MA_1 - MA_2 - MA_3 - MA_4 - MA_5\end{aligned}$$

Subject to:

$$\begin{aligned}&\text{Constraint 1:} && X_1 - S_1 + A_1 = 10000 \\ &\text{Constraint 2:} && X_2 - S_2 + A_2 = 7000 \\ &\text{Constraint 3:} && X_3 - S_3 + A_3 = 5000 \\ &\text{Constraint 4:} && X_4 + S_4 = 4000 \\ &\text{Constraint 5:} && X_5 - S_5 + A_4 = 3000 \\ &\text{Constraint 6:} && X_6 - S_6 + A_5 = 5000 \\ &\text{Constraint 7:} && X_7 + S_7 = 5000 \\ &\text{Constraint 8:} && X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 + S_8 = 50000\end{aligned}$$

Non-negativity Conditions:

$$X_1, X_2, X_3, X_4, X_5, X_6, X_7, S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8, A_1, A_2, A_3, A_4, A_5 \geq 0$$

3.3 Initial Simplex Table

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
$-M$	A_1	1	0	0	0	0	0	0	10000
$-M$	A_2	0	**1**	0	0	0	0	0	7000
$-M$	A_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
$-M$	A_4	0	0	0	0	1	0	0	3000
$-M$	A_5	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	1	1	1	1	1	1	1	50000
	Z	$-M-9$	$-M-10$	$-M-9$	-6	$-M-8$	$-M-10$	-7	$-30000M$

Slack & Artificial Variable Coefficients:

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
A_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
A_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
A_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
A_4	0	0	0	0	-1	0	0	0	0	0	0	1	0
A_5	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	0	0	0	0	0	0	0	1	0	0	0	0	0
Z	M	M	M	0	M	M	0	0	0	0	0	0	0

Enter the variable X_2 and the variable A_2 leaves the base. The pivot element is 1.

3.4 Iterations

Iteration 1

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
$-M$	A_1	1	0	0	0	0	0	0	10000
10	X_2	0	1	0	0	0	0	0	7000
$-M$	A_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
$-M$	A_4	0	0	0	0	1	0	0	3000
$-M$	A_5	0	0	0	0	0	**1**	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	1	0	1	1	1	1	1	43000
	Z	$-M-9$	0	$-M-9$	-6	$-M-8$	$-M-10$	-7	$-23000M+70000$

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
A_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
X_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
A_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
A_4	0	0	0	0	-1	0	0	0	0	0	0	1	0
A_5	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	0	1	0	0	0	0	0	1	0	-1	0	0	0
Z	M	-10	M	0	M	M	0	0	0	$M+10$	0	0	0

Enter the variable X_6 and the variable A_5 leaves the base. The pivot element is 1.

Iteration 2

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
$-M$	A_1	**1**	0	0	0	0	0	-1	10000
10	X_2	0	1	0	0	0	0	0	7000
$-M$	A_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
$-M$	A_4	0	0	0	0	1	0	0	3000
10	X_6	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	1	0	1	1	1	0	1	38000
	Z	$-M-9$	0	$-M-9$	-6	$-M-8$	0	-7	$-18000M+120000$

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
A_1	-1	0	0	0	0	0	0	1	0	0	0	0	0
X_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
A_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
A_4	0	0	0	0	-1	0	0	0	0	0	0	1	0
X_6	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	0	1	0	0	1	0	1	1	0	-1	0	0	-1
Z	M	-10	M	0	M	-10	0	0	0	$M+10$	0	0	$M+10$

Enter the variable X_1 and the variable A_1 leaves the base. The pivot element is 1

Iteration 3

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
9	X_1	1	0	0	0	0	0	0	10000
10	X_2	0	1	0	0	0	0	0	7000
$-M$	A_3	0	0	**1**	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
$-M$	A_4	0	0	0	0	1	0	0	3000
10	X_6	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	0	0	1	1	1	0	1	28000
	Z	0	0	$-M-9$	-6	$-M-8$	0	-7	$-8000M+210000$

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
X_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
X_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
A_3	0	-1	0	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
A_4	0	0	0	0	-1	0	0	0	0	0	0	1	0
X_6	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	1	1	0	0	1	0	1	1	-1	-1	0	0	-1
Z	-9	-10	M	0	M	-10	0	0	$M+9$	$M+10$	0	0	$M+10$

Enter the variable X_3 and the variable A_3 leaves the base. The pivot element is 1.

Iteration 4

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
9	X_1	1	0	0	0	0	0	0	10000
10	X_2	0	1	0	0	0	0	0	7000
9	X_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
$-M$	A_4	0	0	0	0	**1**	0	0	3000
10	X_6	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	0	0	0	1	1	0	1	23000
	Z	0	0	0	-6	$-M-8$	0	-7	$-3000M + 255000$

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
X_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
X_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
X_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
A_4	0	0	0	0	0	0	0	0	0	0	0	1	0
X_6	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	1	1	1	0	1	0	1	1	-1	-1	-1	0	-1
Z	-9	-10	-9	0	M	-10	0	0	$M+9$	$M+10$	$M+9$	0	$M+10$

Enter the variable X_5 and the variable A_4 leaves the base. The pivot element is 1.

Iteration 5

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
9	X_1	1	0	0	0	0	0	0	10000
10	X_2	0	1	0	0	0	0	0	7000
9	X_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
8	X_5	0	0	0	0	1	0	0	3000
10	X_6	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_8	0	0	0	1	0	0	1	20000
	Z	0	0	0	-6	0	0	-7	279000

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
X_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
X_2	0	-1	0	0	0	0	0	0	0	1	0	0	0
X_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
X_5	0	0	0	0	-1	0	0	0	0	0	0	1	0
X_6	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_8	1	**1**	1	0	1	1	0	1	-1	-1	-1	-1	-1
Z	-9	-10	-9	0	-8	-10	0	0	$M+9$	$M+10$	$M+9$	$M+8$	$M+10$

Enter the variable S_2 and the variable S_8 leaves the base. The pivot element is 1.

Iteration 6

C_B	Base	X_1	X_2	X_3	X_4	X_5	X_6	X_7	RHS
9	X_1	1	0	0	0	0	0	0	10000
10	X_2	0	1	0	1	0	0	1	27000
9	X_3	0	0	1	0	0	0	0	5000
0	S_4	0	0	0	1	0	0	0	4000
8	X_5	0	0	0	0	1	0	0	3000
10	X_6	0	0	0	0	0	1	0	5000
0	S_7	0	0	0	0	0	0	1	5000
0	S_2	0	0	0	1	0	0	1	20000
	Z	0	0	0	4	0	0	3	**479000**

Slack & Artificial Variable Coefficients

Base	S_1	S_2	S_3	S_4	S_5	S_6	S_7	S_8	A_1	A_2	A_3	A_4	A_5
X_1	-1	0	0	0	0	0	0	0	1	0	0	0	0
X_2	1	0	1	0	1	1	0	1	-1	0	-1	-1	-1
X_3	0	0	-1	0	0	0	0	0	0	0	1	0	0
S_4	0	0	0	1	0	0	0	0	0	0	0	0	0
X_5	0	0	0	0	0	0	0	0	0	0	0	1	0
X_6	0	0	0	0	0	-1	0	0	0	0	0	0	1
S_7	0	0	0	0	0	0	1	0	0	0	0	0	0
S_2	1	1	1	0	1	1	0	1	-1	-1	-1	-1	-1
Z	1	0	1	0	2	0	0	10	$M-1$	M	$M-1$	$M-2$	M

After completing all iterations using the Big M method, we find that the maximum value of the objective function occurs when the household allocates **Rs. 10,000 on Rent (X_1), Rs. 27,000 on Groceries and Essentials (X_2), Rs. 5,000 on Child's Education (X_3), Rs. 0 on Transport (X_4), Rs. 3,000 on Heathcare (X_5), Rs. 5,000 on Savings (X_6), and Rs. 0 on Entertainment (X_7)**. The total allocation sums to Rs. 50,000, which satisfies the budget constraint $X_1 + X_2 + X_3 + X_4 + X_5 + X_6 + X_7 \leq 50{,}000$, meaning the entire available budget is optimally utilized.

The utility function to be maximized is:

$$Z = 9X_1 + 10X_2 + 9X_3 + 6X_4 + 8X_5 + 10X_6 + 7X_7$$

Substituting the optimal values into the objective function:

$$\begin{aligned} Z &= (9 \times 10{,}000) + (10 \times 27{,}000) + (9 \times 5{,}000) + (6 \times 0) \\ &\quad + (8 \times 3{,}000) + (10 \times 5{,}000) + (7 \times 0) \\ &= 90{,}000 + 270{,}000 + 45{,}000 + 0 + 24{,}000 + 50{,}000 + 0 \\ &= \boxed{479{,}000} \\ &= \mathbf{479\ utils} \end{aligned}$$

Thus, the household can achieve a maximum satisfaction of **479 utils** by allocating its budget as described above. This solution satisfies all constraints and lies within the feasible region, ensuring that the available resources are distributed in a way that maximizes total utility.

3.5 Final Optimal Solution

Optimal Values

Maximum Objective Value: $Z = 479000$

Decision Variables:

$X_1 = 10000$	$X_2 = 27000$	$X_3 = 5000$
$X_4 = 0$	$X_5 = 3000$	$X_6 = 5000$
$X_7 = 0$		

Slack and Surplus Variables:

$S_1 = 0$	$S_2 = 20000$	$S_3 = 0$
$S_4 = 4000$	$S_5 = 0$	$S_6 = 0$
$S_7 = 5000$	$S_8 = 0$	

Artificial Variables:

$A_1 = 0$	$A_2 = 0$	$A_3 = 0$
$A_4 = 0$	$A_5 = 0$	

4 Advantages and Disadvantages of Methods Used

4.1 Graphical Method

The graphical method is ideal for budgeting across **two categories**, such as food and maintainance. It involves drawing a graph to visualize budget constraints and identify the optimal spending split — much like sketching a map of financial choices.

Advantages

- **Easy to Grasp and See:** You can sketch your budget on paper and clearly visualize trade-offs, making it perfect for beginners or small households.
- **Quick to Find the Best Plan:** By sliding the objective line (e.g., $Z = 3x_1 + 2x_2$), the best combination (e.g., $x_1 = 300$, $x_2 = 200$) can be spotted visually without complex calculations.
- **Teaches Budgeting Basics:** It offers a hands-on feel for how constraints like $x_1+x_2 \leq 500$ affect choices and outcomes.
- **Fast for Simple Budgets:** For only two categories and few rules, it can be solved quickly using just a pencil and paper.

Disadvantages

- **Limited to Two Variables:** Cannot handle more than two decision variables, which is impractical for full household budgets.
- **Complex Graphs with Many Constraints:** Too many constraints make the graph cluttered and hard to interpret.
- **Low Precision:** Manual plotting might lead to rounding or drawing errors, especially with decimal values.
- **Hard with Overlapping Constraints:** When constraints intersect oddly, choosing the best solution visually may be unclear.

4.2 Big M Method

The Big M method is designed for **complex budgets** with multiple categories and constraints, including equalities, minimum and maximum spending requirements. It enhances the simplex method by introducing artificial variables with a large penalty term M, ensuring feasibility while seeking optimality.

Advantages

- **Handles Complex Budgets:** Suitable for scenarios with fixed or minimum allocations (e.g., $X_1 \geq$ Rs.10000 for rent).
- **Creates a Starting Point:** Artificial variables provide a feasible initial solution, even when it's not obvious how to start.
- **Reliable and Systematic:** Follows a consistent process to guarantee optimality while obeying all constraints.
- **Great for Larger Households:** Supports many variables like rent, savings, groceries, and utilities simultaneously, especially with computer assistance.

Disadvantages

- **Increases Problem Size:** Adding artificial and surplus variables makes the formulation longer and more complex.
- **Sensitive to Big M:** Improper handling of M in software can cause numerical instability or wrong results.
- **Requires Math Proficiency:** Needs understanding of matrix operations and pivoting, making it less beginner-friendly.
- **Time-Consuming Manually:** Without a computer, solving big problems manually may involve many tableau iterations.

4.3 When to Use Each Method

- **Graphical Method:** Use this for small, two-variable budgets (e.g., food and maintainance). It is quick, visual, and intuitive, making it ideal for individuals or small households just starting with budgeting.
- **Big M Method:** Use this for full household budgeting with multiple needs and conditions (e.g., fixed rent, savings goals, child's education). It ensures all constraints are respected and finds the best allocation, especially effective when paired with software.

5 Conclusion

The application of Linear Programming in household budget allocation illustrates how mathematical optimization can aid in making efficient financial decisions. Through this project, we observed that by properly formulating the problem with decision variables, objective functions, and constraints, it is possible to identify an optimal spending plan that maximizes household satisfaction within a fixed budget. The graphical method provided an intuitive solution for two-variable problems, while the Big M method enabled solving more complex multi-variable problems. This experience highlights the practical utility of LPP in everyday life and reinforces the importance of optimization techniques in resource management. Overall, this project enhanced our understanding of LPP concepts, solution methods, and their real-world applications.